Supporting Information

Ultrafast recombination and exchange coupling preclude weak-field magnetosensitivity of neurotransmitter-metabolising flavoenzymes

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S1 Validation of the Liouville-space solver

The time-integrated singlet yield (Eq. 3 of the main text) is evaluated by exact linear solution in the 32^2-dimensional Liouville space. As a correctness check, in the limit of zero electronic dephasing ($1/T_2^e \to 0$) and symmetric recombination ($k_S = k_T$) the master-equation result must coincide with the closed-form coherent singlet yield obtained from the eigenbasis Green's function,

$$\Phi_S(B) = k \sum_{m,n} \frac{[\tilde{P}_S]_{mn}\,[\tilde{\rho}_0]_{nm}}{k + i(E_m - E_n)}, \tag{S1}$$

where tildes denote the Hamiltonian eigenbasis. The two evaluations agree to machine precision (absolute difference $< 10^{-9}$ across all fields tested); this test is included in the released test suite (`test_master_equation.py`).

S2 The reported value is physical, not a numerical floor

A response of order $10^{-12}\,\%$ evaluated in double precision invites the objection that it is a rounding artefact of the 1024-dimensional linear solve rather than a property of the model. It is not, and the check is independent of the solver's own conventions: in the perturbative regime $\omega_B \ll 1/\tau$ the field enters the Hamiltonian only through ω_B and time-reversal symmetry forbids a linear term, so the MFE must scale as B^2. A numerical floor would not.

Table S1 reports $\mathrm{MFE}(B)/B^2$ for the MAO-A parameter set over three decades of field. From $100\,\mu$T to 25 mT the ratio is constant to better than 0.3 %, in both the coherent limit and with $T_2^e = 1$ ns; the geomagnetic point at $50\,\mu$T lies 1.4 % (coherent) and 0.6 % (realistic) below the converged slope, which sets the accuracy of the quoted values. Extrapolating the converged slope back to $50\,\mu$T gives $1.71 \times 10^{-12}\,\%$ and $1.65 \times 10^{-12}\,\%$ respectively, confirming Table 1 of the main text. Only below $\sim 25\,\mu$T does the double-precision floor intrude and the scaling break down.

S3 The null result is independent of the hyperfine assignment

Because the flavin-radical hyperfine couplings enter only through the singlet–triplet mixing that the exchange coupling and finite lifetime already suppress, the verdict for the active-site pairs does not depend on the precise hyperfine values used. Table S2 lists the calculated geomagnetic-field MFE for the monoamine-oxidase regime ($J = 750$ MHz, $\tau \approx 10$ ps) as the dominant flavin hyperfine constant is varied over more than an order of magnitude: the effect stays at the $10^{-12}\,\%$ level throughout. For the spatially separated cryptochrome-like geometry the MFE

Table S1: Verification that the computed MFE is physical. MAO-A parameters ($J = 750$ MHz, $D = -1214$ MHz, $\tau = 10$ ps). A constant MFE/B^2 is the signature of the perturbative B^2 response; a numerical floor would show none.

	coherent ($T_2^e \to \infty$)		realistic ($T_2^e = 1$ ns)	
B	MFE (%)	MFE/B^2 (10^{-16} %/μT^2)	MFE (%)	MFE/B^2 (10^{-16} %/μT^2)
25 μT	4.00×10^{-13}	6.395	4.04×10^{-13}	6.458
50 μT	1.69×10^{-12}	6.750	1.64×10^{-12}	6.548
100 μT	6.84×10^{-12}	6.839	6.58×10^{-12}	6.582
500 μT	1.71×10^{-10}	6.848	1.65×10^{-10}	6.583
1 mT	6.85×10^{-10}	6.849	6.58×10^{-10}	6.584
5 mT	1.71×10^{-8}	6.848	1.65×10^{-8}	6.583
25 mT	4.27×10^{-7}	6.836	4.11×10^{-7}	6.571

varies within a factor of about four over the same range (+0.26% to +1.07%), confirming that the twelve-order-of-magnitude contrast between the two regimes is a property of the kinetics, the exchange and dipolar couplings, and not of the hyperfine model.

Table S2: Calculated MFE at $B = 50\,\mu$T as the dominant flavin hyperfine constant A is varied. "Active-site" uses $J = 750$ MHz, $\tau \approx 10$ ps, $T_2^e = 1$ ns (the monoamine-oxidase regime); "separated" uses $J \approx 0$, $\tau = 1\,\mu$s, $T_2^e = 1\,\mu$s (the cryptochrome regime).

A_{P} (MHz)	MFE, active-site (%)	MFE, separated (%)
5	1.50×10^{-12}	+0.91
10	1.59×10^{-12}	+1.07
20	1.77×10^{-12}	+0.86
50	3.32×10^{-12}	+0.67
100	8.80×10^{-12}	+0.26
200	3.08×10^{-11}	+0.26

S4 What controls the bound

The physical content of the result is the separation of scales between the two regimes, summarised here for reference:

- **Exchange coupling.** Weak-field singlet–triplet mixing requires the singlet and triplet manifolds to be near-degenerate on the scale of the Zeeman and hyperfine energies ($\lesssim$ tens of MHz). At an active-site inter-radical distance ($r \approx 3.5$ Å), $J \sim 0.4$–0.75 GHz holds them apart by one to two orders of magnitude, so a 50 μT field cannot bring them into resonance.

- **Lifetime.** Coherent mixing driven by sub-millitesla hyperfine fields develops over hundreds of nanoseconds. Picosecond back electron transfer ($\tau \approx 10$ ps) freezes the spin state long before mixing occurs.

- **Electronic decoherence.** Hyperfine-induced dephasing at the active site ($T_2^e \sim 1$ ns) damps any residual coherence two orders of magnitude faster than the $\gtrsim 100$ ns required for an effect to accumulate.

The picosecond lifetime is the mechanism that robustly enforces the bound; the active-site radical pairs of the neurotransmitter-metabolising flavoenzymes are subject to all three.

S5 Robustness to the quantum Zeno mechanism

A tightly bound radical pair can become sensitive to a weak field through a quantum Zeno effect if its recombination is strongly spin-asymmetric. We tested whether this rescues the monoamine-oxidase regime by recomputing the MFE for $|J| = 750$ MHz, $\tau \approx 10$ ps while varying the singlet/triplet recombination asymmetry k_S/k_T over six orders of magnitude (Table S3), both with electronic dephasing ($T_2^e = 1$ ns) and without. At the geomagnetic field the MFE stays below 10^{-7} % for every asymmetry; even at a 200-fold stronger field (10 mT) it does not exceed 1.5×10^{-3} %. The Zeno mechanism cannot open a magnetosensitive window because the picosecond active-site lifetime quenches the spin evolution before any asymmetry in recombination can project it.

Table S3: Calculated MFE for the monoamine-oxidase regime ($|J| = 750$ MHz, $\tau \approx 10$ ps, $T_2^e = 1$ ns) as the recombination asymmetry k_S/k_T is varied. Even strongly spin-asymmetric recombination leaves the geomagnetic-field effect at zero.

k_S/k_T	τ_T	MFE (50 μT)	MFE (1 mT)	MFE (10 mT)
1	10 ps	$+1.6 \times 10^{-12}$	$+6.6 \times 10^{-10}$	$+6.6 \times 10^{-8}$
10^2	1 ns	-5.0×10^{-9}	-2.0×10^{-6}	-2.2×10^{-4}
10^4	100 ns	-1.2×10^{-8}	-4.7×10^{-6}	-5.4×10^{-4}
10^6	10 μs	-6.6×10^{-11}	-2.6×10^{-8}	-2.3×10^{-6}

S6 Robustness to singlet–triplet dephasing

For a tightly bound radical pair the dominant electronic decoherence is expected to be pure dephasing between the singlet and triplet manifolds, driven by fluctuations in the large exchange coupling $2J$, rather than the single-spin "random-field" (S_z) channel used in the main text. The two channels have different Lindblad operators: the $2J$-fluctuation channel is generated by $A = P_S - P_T$ (since the fluctuating term of the Hamiltonian is $\propto P_S - P_T$), giving the dissipator $\gamma_{ST}\,(A\rho A - \rho)$, which damps singlet–triplet coherence at rate $2\gamma_{ST}$ without affecting populations.

We added this channel to the monoamine-oxidase model ($|J| = 750$ MHz, $\tau \approx 10$ ps, with the S_z channel also present at $T_2^e = 1$ ns) and varied γ_{ST} from zero up to 10^4 MHz—comparable to and beyond the exchange coupling itself (Table S4). The geomagnetic-field MFE remains at the 10^{-12} % level throughout, and even at a 200-fold stronger field (10 mT) it stays at $\sim 4 \times 10^{-11}$ %. This is expected: because the active-site effect already vanishes in the fully coherent limit (Table S1), the additional dephasing has little left to act on. We do not claim this as a general theorem: because the MFE is a ratio, a dissipative channel can in principle increase it, and the main text gives an explicit counterexample for a long-lived pair. The statement here is empirical and restricted to the two channels and the picosecond regime examined.

S7 Robustness to the Δg mechanism

The one coherent channel that survives when the exchange coupling greatly exceeds the hyperfine scale is the Δg mechanism, in which the two radicals precess at different rates because their g-factors differ. Because it does not rely on hyperfine coupling, it is the natural candidate for restoring sensitivity in a strongly exchange-coupled pair, and we therefore tested it explicitly. We added the differential Zeeman term

$$H_{\Delta g} = \frac{\Delta g\, \mu_B B}{\hbar}\,(S_{1z} - S_{2z}) \tag{S2}$$

Table S4: Calculated MFE for the monoamine-oxidase regime ($|J| = 750$ MHz, $\tau \approx 10$ ps, $T_2^e = 1$ ns) as a singlet–triplet dephasing channel $A = P_S - P_T$ of rate γ_{ST} (from $2J$ fluctuations) is added. The null is unaffected for any rate up to and beyond the exchange coupling.

γ_{ST} (MHz)	MFE (50 μT) (%)	MFE (10 mT) (%)
0	1.64×10^{-12}	6.58×10^{-8}
10^1	1.64×10^{-12}	6.56×10^{-8}
10^2	1.60×10^{-12}	6.34×10^{-8}
10^3	1.17×10^{-12}	4.64×10^{-8}
10^4	1.57×10^{-13}	5.86×10^{-9}

to Eq. (1) and recomputed the monoamine-oxidase case (Table S5). At $\Delta g \simeq 10^{-3}$, the realistic value for a pair of organic radicals, the geomagnetic-field MFE is essentially unchanged (it in fact decreases slightly). Only at $\Delta g = 3 \times 10^{-2}$—more than an order of magnitude larger than any organic radical pair—does the effect reach 3×10^{-10} %, still eight orders of magnitude below any measurable level.

Table S5: Calculated MFE at $B = 50\,\mu$T for the monoamine-oxidase regime ($J = 750$ MHz, $D = -1214$ MHz, $\tau \approx 10$ ps, $T_2^e = 1$ ns) as a Δg term is added. Realistic organic-radical values are $\Delta g \lesssim 10^{-3}$.

Δg	MFE (50 μT) (%)	ratio to $\Delta g = 0$
0	$+1.64 \times 10^{-12}$	1.00
10^{-4}	$+1.64 \times 10^{-12}$	1.00
10^{-3}	$+1.27 \times 10^{-12}$	0.77
3×10^{-3}	-1.65×10^{-12}	-1.01
10^{-2}	-3.49×10^{-11}	-21.3
3×10^{-2}	-3.28×10^{-10}	-200

Why no coherent mechanism rescues the picosecond pair

This result is a particular case of a general scale argument. The couplings in the problem order as

$$\underbrace{1/\tau = 10^5}_{\text{lifetime}} \gg \underbrace{|D| = 1214}_{\text{dipolar}} > \underbrace{J = 750}_{\text{exchange}} \gg \underbrace{A \simeq 44}_{\text{hyperfine}} \gg \underbrace{\omega_B \simeq 1.4}_{\text{Zeeman at } 50\,\mu\text{T}} \tag{S3}$$

in MHz. The recombination rate exceeds the largest internal coupling by more than two orders of magnitude, so the pair is destroyed before any coherent term can act; this is why the active-site and separated geometries give the same MFE to within a few per cent at $\tau = 10$ ps (Fig. 1a), despite differing by three orders of magnitude in J. More fundamentally, the magnetic field enters the Hamiltonian only through ω_B, so at leading order the response is bounded by $(\omega_B \tau)^2 \approx 7.8 \times 10^{-9}$ as a dimensionless fraction, i.e. 7.8×10^{-7} %, irrespective of how the remaining couplings are arranged—comfortably above the computed 1.6×10^{-12} %, whose B^2 scaling is verified over more than two decades of field in Table S1. Refinements to the internal Hamiltonian (anisotropic hyperfine and g tensors, the correct ^{14}N nuclear spin of 1, substrate-specific couplings) change the prefactor, not this ceiling.

S8 Computational details for the electronic-structure inputs

S8.1 Geometries and hyperfine/distance inputs (GFN2-xTB)

Active-site geometries were built from the crystallographically resolved monoamine-oxidase structure [1] and a reduced lumiflavin–methylamine model of the MAO-A active site (the isoalloxazine ring of FAD truncated at N10, plus methylamine as the substrate amine). Geometries were optimised with the semi-empirical tight-binding method GFN2-xTB [2] (neutral, closed-shell; `xtb --opt --chrg 0 --uhf 0`). The hydrogen-transfer coordinate was sampled by constraining the $d(\mathrm{N5}\cdots\mathrm{H})$ distance over 1.0–1.8 Å and re-optimising the remaining degrees of freedom. Inter-radical distances and the dominant isotropic hyperfine magnitudes used in the spin Hamiltonian were read from these structures and cross-checked against published ENDOR hyperfine couplings for protein flavin radicals [3]; the spin-dynamics conclusions are independent of the precise hyperfine values (table S1 above).

S8.2 Multireference diradical character (CASSCF/NEVPT2)

At each scan point the diradical character of the singlet was evaluated with complete-active-space self-consistent field theory in the def2-SVP basis, CASSCF(2,2)/def2-SVP, as implemented in PySCF [4]. The active space comprises the two frontier orbitals involved in a putative single-electron transfer (the flavin LUMO and the amine-nitrogen lone-pair HOMO); this is the minimal active space that can resolve closed-shell versus diradical singlet configurations. Dynamic correlation was added by strongly contracted n-electron valence-state perturbation theory (NEVPT2) [5] on the CASSCF reference. The lowest singlet and triplet were computed at each geometry. The diradical index was taken as $y_0 = 2\,c_{02}^2$ from the weight of the doubly-excited configuration in the CASSCF singlet wavefunction [6], equivalent to the occupation of the lowest unoccupied natural orbital. Table S6 collects the results.

Table S6: CASSCF(2,2)/def2-SVP diradical index y_0 and CASSCF singlet–triplet gap along the hydrogen-transfer coordinate of the lumiflavin–methylamine model. Small y_0 and a large S–T gap indicate a predominantly closed-shell (polar/concerted) singlet, not a spin-correlated radical pair.

$d(\mathrm{N5}\cdots\mathrm{H})$ (Å)	y_0	S–T gap (eV)
1.0	0.032	3.24
1.4	0.092	4.10
1.8	0.091	3.61

The diradical index stays below 0.10 across the coordinate, with a singlet–triplet gap of 3–4 eV—two orders of magnitude larger than any hyperfine or Zeeman energy. CAS(2,2) is the minimal frontier active space; larger active spaces (e.g. CAS(4,4)/(6,6) including additional flavin π orbitals) would refine the absolute numbers, but the qualitative conclusion—weak diradical character and a large S–T gap, i.e. no genuine radical-pair intermediate—is robust across the scanned geometries and is consistent with QM/MM mechanistic studies of MAO [7, 8].

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